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# DECISION TREE CLASSIFIER

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from sklearn.datasets import load_iris
from sklearn import tree
import matplotlib.pyplot as plt

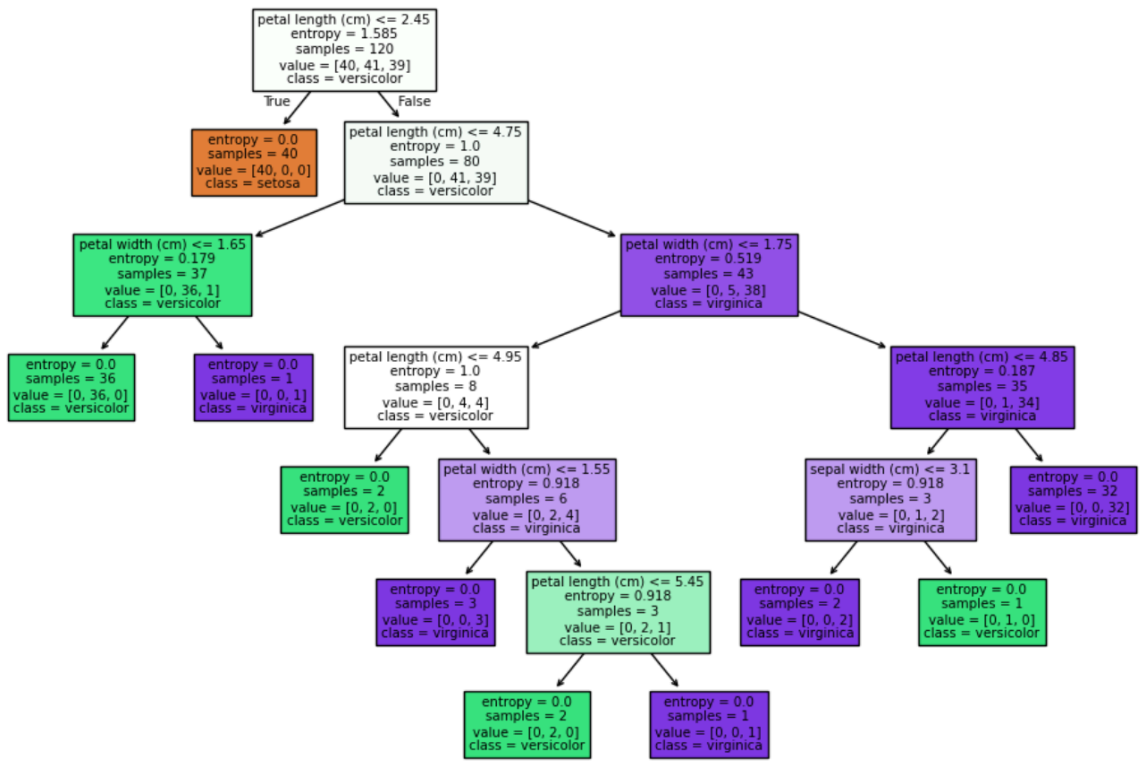
# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Train Test Split (80/20)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Decision Tree Model
model = DecisionTreeClassifier(criterion="entropy", random_state=42)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Results
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))

# Visualize Tree
plt.figure(figsize=(12,8))
tree.plot_tree(model, feature_names=iris.feature_names,
               class_names=iris.target_names,
               filled=True)
plt.show()
```

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Accuracy: 1.0
Confusion Matrix:
 [[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         9
           2       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```